# Continuous-Symbol Hankel Compactness on Product Domains

Partial solution packet for arXiv:2011.02656

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**Status: candidate substantial partial theorem, likely valid.** Remark 2 of the source has an affirmative answer, at every form level, on products of two bounded strictly convex domains. It also has an affirmative $q = 0$ answer on arbitrary finite products of bounded convex domains. The general convex-domain question remains open.

## 1 Source question and result

Let $\Omega \subset \mathbb{C}^n$ be a bounded convex domain. For $0 \le q \le n-1$, write $K^2_{(0,q)}(\Omega)$ for the square-integrable $\overline{\partial}$-closed $(0,q)$-forms, $P_q$ for the orthogonal projection onto that space, and

$$H^q_\varphi f = (I - P_q)(\varphi f).$$

Celik, Sahutoglu, and Straube prove that if $\varphi \in C^1(\overline{\Omega})$ is holomorphic along all analytic varieties in $\partial\Omega$ of dimension at least $q+1$, then $H^q_\varphi$ is compact [1, Theorem 1]. Their Remark 2 asks whether $C^1(\overline{\Omega})$ can be replaced by $C(\overline{\Omega})$.

Our main result settles that question on a product class with uncountably many side varieties.

**Theorem 1** (Two strictly convex factors)**.** *Let $D \subset \mathbb{C}^a$ and $G \subset \mathbb{C}^b$ be bounded strictly convex domains, $\Omega = D \times G$, and $0 \le q \le a+b-1$. Suppose $\varphi \in C(\overline{\Omega})$ is holomorphic along every analytic variety in $\partial\Omega$ of dimension at least $q+1$. Then*

$$H^q_\varphi : K^2_{(0,q)}(\Omega) \longrightarrow L^2_{(0,q)}(\Omega)$$

*is compact.*

For $q = 0$, strict convexity of the factors is unnecessary and any finite number of factors is allowed; see Corollary 1.

## 2 Two approximation principles

We repeatedly use the elementary operator bound

$$\|H^q_\varphi - H^q_\psi\| = \|(I - P_q)M_{\varphi-\psi}|_{K^2_{(0,q)}}\| \le \|\varphi - \psi\|_\infty. \tag{1}$$

Thus a uniform limit of symbols with compact Hankel operators again has a compact Hankel operator.

For a bounded domain $U$, put

$$A(U) = C(\overline{U}) \cap \mathcal{O}(U)$$

with the uniform norm. The following boundary-cross lemma is the central device.

**Theorem 1.** *Let $\Omega$ be a bounded convex domain in $\mathbb{C}^n$, $\varphi \in C^1(\overline{\Omega})$, and $0 \le q \le n-1$. Assume that $\varphi$ is holomorphic along the varieties in the boundary of dimension $q+1$ (and higher). Then $H^q_\varphi$ is compact on $K^2_{(0,q)}(\Omega)$.*

Combining Theorem 1 with [ÇŞS20, Theorem 1], we have a complete characterization of compactness of $H^q_\varphi$ for symbols in $C^1(\overline{\Omega})$.

**Corollary 1.** *Let $\Omega$ be a bounded convex domain in $\mathbb{C}^n$, $\varphi \in C^1(\overline{\Omega})$, and $0 \le q \le n-1$. Then $H^q_\varphi$ is compact if and only if $\varphi$ is holomorphic along complex varieties of dimension $(q+1)$ (and higher) in the boundary.*

*Remark* 1. Theorem 1 in [ÇŞS20] only assumes that the Hankel operator is compact on $A^2_{(0,q)}(\Omega)$. Thus, on a bounded convex domain, if the symbol $\varphi$ is in $C^1(\overline{\Omega})$, then $H^q_\varphi$ is compact on $K^2_{(0,q)}(\Omega)$ if and only if it is compact on $A^2_{(0,q)}(\Omega)$. The analogous fact holds for the $\overline{\partial}$–Neumann operator, see [FS98]. For related questions, see [Has14, Theorem 12.12].

*Remark* 2. The converse of Theorem 1, [ÇŞS20, Theorem 1], holds for $\varphi$ only assumed in $C(\overline{\Omega})$. We do not know whether the conclusion of Theorem 1 still holds under this weaker assumption on $\varphi$. One natural approach would be to approximate $\varphi \in C(\overline{\Omega})$ by suitable symbols in $C^1(\overline{\Omega})$. Our proof below indicates that it suffices to approximate $\varphi \in C(\overline{\Omega})$, holomorphic along the varieties in the boundary, uniformly on $\overline{\Omega}$ to within $\varepsilon$ by $\varphi_\varepsilon \in C^1(\overline{\Omega})$ with $|\overline{\partial}_V \varphi_\varepsilon| \le \eta(\varepsilon)$, where $\lim_{\varepsilon\to 0^+} \eta(\varepsilon) = 0$ (rather than with $\overline{\partial}_V \varphi_\varepsilon = 0$). Here, $\overline{\partial}_V$ denotes $\overline{\partial}_V$ along $V \subset b\Omega$.

Figure 1: The source theorem and open continuous-symbol question on source page 2. The displayed page was reconstructed from the run's ingested official arXiv source.

**Lemma 1** (Boundary cross)**.** *Let $D \subset \mathbb{C}^a$ and $G \subset \mathbb{C}^b$ be bounded domains and let $\varphi \in C(\overline{D} \times \overline{G})$. Assume*

$$\begin{aligned} w &\longmapsto \varphi(\xi, w) \in A(G) \quad (\xi \in \partial D), \\ z &\longmapsto \varphi(z, \eta) \in A(D) \quad (\eta \in \partial G). \end{aligned}$$

*Then there is a unique $h \in A(D \times G)$ satisfying $h = \varphi$ on $\partial(D \times G)$.*

*Proof.* Let $B_D \subset C(\partial D)$ be the boundary traces of $A(D)$. Restriction $T_D : A(D) \to B_D$ is an isometry by the maximum principle. Hence $B_D$ is closed and $R_D = T_D^{-1}$ is an isometry.

Define $F : \overline{G} \to C(\partial D)$ by $F(w)(\xi) = \varphi(\xi, w)$. Uniform continuity of $\varphi$ makes $F$ continuous. It is Banach-valued holomorphic on $G$: for every finite measure $\mu$ on $\partial D$, Morera's theorem and Fubini show that $w \mapsto \int_{\partial D} \varphi(\xi, w)\, d\mu(\xi)$ is holomorphic.

If $\Lambda \in C(\partial D)^*$ annihilates $B_D$, then $\Lambda(F(\cdot)) \in A(G)$. For $\eta \in \partial G$ the second slice assumption gives $F(\eta) \in B_D$, so this scalar function vanishes on $\partial G$. The maximum principle gives $\Lambda(F(w)) = 0$ on $G$. Hahn–Banach now implies $F(w) \in B_D$ for every $w \in \overline{G}$.

Set $H(w) = R_D F(w) \in A(D)$ and $h(z, w) = H(w)(z)$. The map $H$ is continuous on $\overline{G}$ and Banach-holomorphic on $G$. Thus $h$ is separately holomorphic, hence jointly holomorphic by Hartogs' theorem, and is continuous on the closed product. Its trace agrees with $\varphi$ on $\partial D \times \overline{G}$ by construction. On $\overline{D} \times \partial G$, the two $A(D)$ functions have the same boundary trace and hence agree. Uniqueness follows from the maximum principle. $\square$

We also need a one-sided version at the level of approximation.

**Lemma 2** (One-sided smooth approximation). *Let $D \subset \mathbb{C}^a$ be bounded and convex, $G \subset \mathbb{C}^b$ bounded, and $h \in C(\overline{D} \times \overline{G})$ be holomorphic in $z \in D$ for each $w \in \overline{G}$. Then $h$ is a uniform limit of functions in $C^1(\overline{D} \times \overline{G})$ that are holomorphic in $z$.*

*If $g \in C(\overline{D} \times \overline{G})$ vanishes on $\overline{D} \times \partial G$, then it is a uniform limit of $C^1$ functions that vanish on a neighborhood of that side.*

*Proof.* Regard $h$ as a continuous map $\overline{G} \to A(D)$. Polynomials are dense in $A(D)$: dilate toward an interior point of the convex domain and then apply Oka–Weil. The compact image $h(\overline{G})$ therefore has a finite cover by small balls centered at polynomials $p_j$. A smooth partition of unity $\chi_j$ on a neighborhood of $\overline{G}$, subordinate to the inverse-image cover, gives

$$h_\varepsilon(z, w) = \sum_j \chi_j(w) p_j(z),$$

with $\|h - h_\varepsilon\|_\infty < \varepsilon$.

For the second assertion, uniform continuity and vanishing on the indicated side allow multiplication by a continuous cutoff in $w$ that is zero near $\partial G$ with arbitrarily small uniform error. Approximate the resulting continuous function uniformly on the compact product by smooth functions (real-coordinate polynomials suffice), and multiply by a slightly larger smooth cutoff that is identically one on the first cutoff's support. The approximants remain zero near $\overline{D} \times \partial G$. □

The same cutoff argument shows that a continuous function vanishing on the full boundary of a bounded domain is uniformly approximable by $C^1$ functions supported away from the boundary. Also, every continuous function on a compact subset of Euclidean space is uniformly approximable by smooth functions.

## 3 Boundary geometry of a strict product

**Lemma 3.** *Every positive-dimensional analytic boundary variety of $D \times G$, with $D$ and $G$ strictly convex, is contained in a side*

$$\{\xi\} \times \overline{G} \quad (\xi \in \partial D), \qquad \text{or} \qquad \overline{D} \times \{\eta\} \quad (\eta \in \partial G).$$

*Consequently its dimension is at most $b$ in the first case and at most $a$ in the second.*

*Proof.* Consider a connected irreducible local parameterization $v = (v_D, v_G)$ of the variety. Both coordinate maps take values in the corresponding closed domains. If $v_D$ reaches a point of $\partial D$, choose a real supporting hyperplane there. Its complex-linear defining functional, composed with $v_D$, has real part bounded above and attaining its maximum at an interior parameter point. The maximum principle places the entire image of $v_D$ in that supporting face. Strict convexity makes the face a singleton, so $v_D$ is constant at a boundary point. If $v_D$ never reaches $\partial D$, then $v_D$ lies in $D$; because $v$ lies in $\partial(D \times G)$, $v_G$ lies in $\partial G$ everywhere, and the symmetric argument makes $v_G$ constant. Applying this componentwise proves the claim and the dimension bounds. □

This argument is also consistent with the source's observation that boundary varieties of convex domains are affine intersections [1, 2].

## 4 Proof of the main theorem

*Proof of Theorem 1.* There are three cases.

*Case 1: $a, b \geq q+1$.* For every $\xi \in \partial D$, the side $\{\xi\} \times G$ is a $b$-dimensional boundary variety, so $\varphi(\xi, \cdot) \in A(G)$. Similarly $\varphi(\cdot, \eta) \in A(D)$ for every $\eta \in \partial G$. Lemma 1 supplies $h \in A(D \times G)$ agreeing with $\varphi$ on the full boundary. Put $g = \varphi - h$. Then $H^q_h = 0$, because multiplication by a holomorphic function preserves $\overline{\partial}$-closedness. Approximate $g$ uniformly by $g_\nu \in C^1(\overline{\Omega})$ supported away from $\partial\Omega$. Each $g_\nu$ is zero along every boundary variety, so the source theorem makes $H^q_{g_\nu}$ compact. Equation (1) gives compactness of $H^q_g$ and hence of $H^q_\varphi$.

*Case 2: exactly one of $a, b$ is at least $q+1$.* Assume $a \geq q+1 > b$; the other case is symmetric. Lemma 3 shows that every boundary variety of dimension at least $q+1$ lies in $D \times \{\eta\}$ for some $\eta \in \partial G$. Hence

$$F(\eta) = \varphi(\cdot, \eta) \in A(D), \qquad \eta \in \partial G,$$

and $F : \partial G \to A(D)$ is continuous.

Extend $F$ continuously to $\overline{G}$ with values in $A(D)$. For completeness, one explicit extension is obtained by choosing $w_0 \in G$ and an $F_0 \in A(D)$, then interpolating linearly in $A(D)$ from $F_0$ to the boundary value along each ray from $w_0$; convexity of $G$ makes the radial gauge continuous. Denote the associated scalar function by $h(z, w)$. It is continuous on the closed product, holomorphic in $z$, and agrees with $\varphi$ on $\overline{D} \times \partial G$.

By Lemma 2, choose $C^1$ functions $h_\nu$, holomorphic in $z$, with $h_\nu \to h$ uniformly. The remainder $g = \varphi - h$ vanishes on $\overline{D} \times \partial G$; choose $C^1$ functions $g_\nu \to g$ uniformly that vanish near that side. Both $h_\nu$ and $g_\nu$ are holomorphic along every boundary variety of dimension at least $q+1$: the former are holomorphic in the only possible varying coordinate, and the latter are zero there. The source theorem makes $H^q_{h_\nu}$ and $H^q_{g_\nu}$ compact. Equation (1), applied to their sums, proves compactness of $H^q_\varphi$.

*Case 3: $a, b \leq q$.* Lemma 3 says there are no boundary varieties of dimension at least $q+1$. Approximate $\varphi$ uniformly by arbitrary $C^1$ functions. The source hypothesis is vacuous for every approximant, so the source theorem and (1) finish the proof. □

## 5 A finite-product corollary

**Corollary 1.** *Let $\Omega = \Omega_1 \times \cdots \times \Omega_m$, $m \geq 2$, where every $\Omega_j$ is a bounded convex domain of positive dimension. If $\varphi \in C(\overline{\Omega})$ is holomorphic along every positive-dimensional analytic variety in $\partial\Omega$, then the scalar Bergman-space Hankel operator $H^0_\varphi$ is compact.*

*Proof.* Each side obtained by fixing one coordinate at a boundary point is a positive-dimensional boundary variety, so the restriction of $\varphi$ to it is holomorphic. Group $\Omega_1$ as $D$ and the product of the remaining factors as $G$. The two slice conditions of Lemma 1 hold: if a point of $\partial G$ has several boundary coordinates, any one of them places the entire $D$-slice in a side face. Thus $\varphi = h + g$ with $h \in A(\Omega)$ and $g = 0$ on $\partial\Omega$. Now $H^0_h = 0$, and smooth interior-supported approximation of $g$, followed by the source theorem and (1), proves the claim. □

## 6 Checks, novelty, and limitation

**Logical checks.** The form-level proof never asserts that multiplication by an interior-supported function is compact on $K^2_{(0,q)}$; that assertion would be false in general for $q > 0$. Instead every smooth approximant is fed into the source theorem. The only closure step is the elementary operator-norm estimate (1). The boundary-cross argument uses no boundary regularity.

**Bounded novelty search.** The run indexes and bounded title/keyword searches found the earlier theorem for finitely many disjoint boundary varieties [2], results on products of Hankel operators on the polydisc, and continuous-symbol results for convex Reinhardt domains under additional hypotheses. No theorem matching Theorem 1 or Corollary 1 was found. The present product has uncountably many side varieties, so it is not covered by the finite-variety theorem. Novelty is plausible, not certified.

**Limitation.** The packet does not solve Remark 2 for an arbitrary bounded convex domain. Without strict product geometry, affine boundary faces can be nested and accumulate, and the unresolved step is a compatible smooth approximation across that entire face system.

## References


[1] M. Celik, S. Sahutoglu, and E. J. Straube, *A sufficient condition for compactness of Hankel operators*, arXiv:2011.02656 (2020), https://arxiv.org/abs/2011.02656.

[2] M. Celik, S. Sahutoglu, and E. J. Straube, *Compactness of Hankel operators with continuous symbols on convex domains*, Houston J. Math. **46** (2020), no. 4, 1005–1016; arXiv:2005.14323.